Loss Functions as Noise Models

Signal Detection and Estimation Theory, Project 3

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The problem

- Each loss function is the negative log-likelihood of one noise model.
- MSE assumes Gaussian noise. MAE assumes Laplace noise.
- **Question:** what is the cost when the noise model is wrong?

Model. We estimate a constant A from N samples:

$$x[n] = A + w[n], \qquad n = 0, \ldots, N-1, \qquad w[n] \text{ i.i.d., zero mean.}$$

This is linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \mathbf{w}$ with $\mathbf{X} = \mathbf{1}$.

Measure of cost.

$$\text{efficiency ratio} = \frac{\text{Var}(\hat{A})}{\text{CRLB}}$$

A ratio of 2 means twice the variance of the best estimator. If both variances fall as 1/N, the estimator needs 2 times more data. If the variance stops falling, more data will not help.

The loss comes from the noise model

Gaussian noise

$$\ln p(\mathbf{x}; A) = c - \frac{1}{2\sigma^2} \sum_n (x[n] - A)^2$$

The MLE minimises the squared error.

$$\Rightarrow \hat{A} = \bar{x} \quad \text{(mean)}$$

Laplace noise

$$\ln p(\mathbf{x}; A) = c - \frac{1}{b} \sum_n |x[n] - A|$$

The MLE minimises the absolute error.

$$\Rightarrow \hat{A} = \text{median}(\mathbf{x})$$

- For the median, the derivative of $\sum |x - A|$ is (samples below A) $-$ (samples above A). It changes sign at the middle sample.
- Choosing a loss is thus to choose a noise model!

CRLB

- Gaussian: σ^2/N
- Laplace: b^2/N

Variance of each estimator

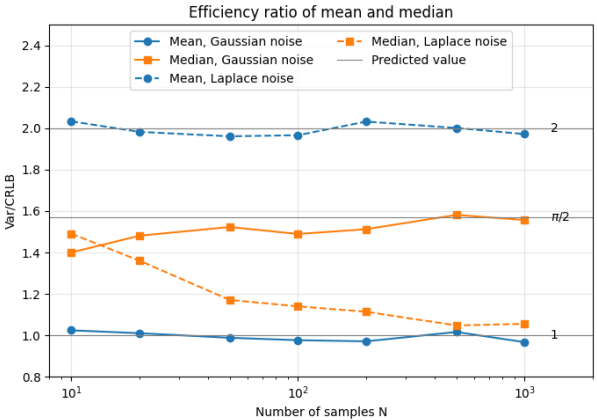
- Mean: $\mathrm{Var}(w)/N$ (exact for all N)
- Median: $\approx 1/(4N\,p_w(0)^2)$ (large N)

Predicted efficiency ratio

	Gaussian	Laplace
Mean (MSE)	1	2
Median (MAE)	$\pi/2$	1

- Each estimator is best only for its own noise.
- Under Laplace noise, the mean needs 2 times more data than the median.

Simulation with Gaussian and Laplace noise

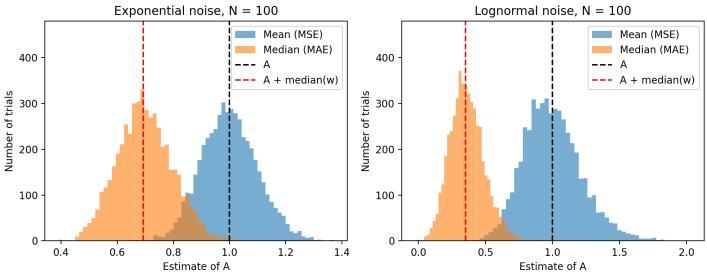


5000 trials. Measured at $N = 100$:

	Gauss.	Laplace
Mean	0.97	2.01
Median	1.51	1.17

- The mean is flat for all N.
- The median reaches its value only for large N.
- At small N, our median results match exact published values (NIST, 1964).

Skewed noise: the loss changes the target



Consider the case of zero-mean noise, but now not symmetric!

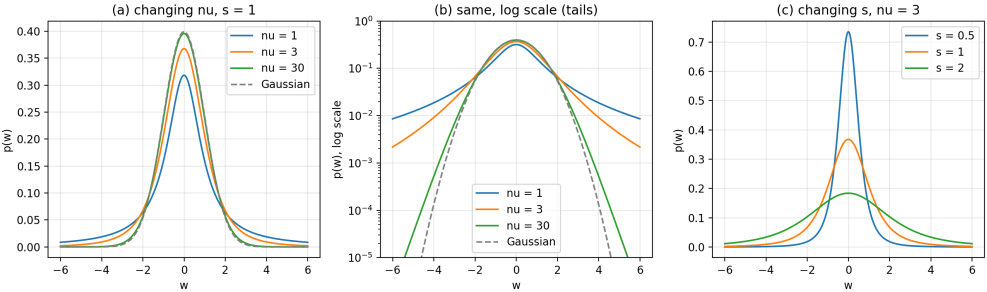
- Mean $= A$
- Median $= A + \text{median}(w)$

	Pred.	Meas.
Expon.	0.693	0.698
Lognorm.	0.351	0.363

MSE is not biased. MSE and MAE estimate different values.

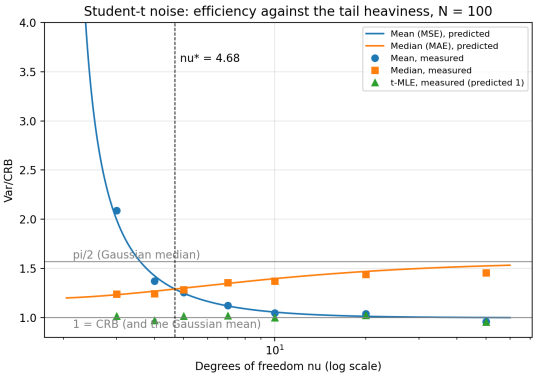
Student-t noise: the model

$$p_w(w) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)s}\left(1+\frac{w^2}{\nu s^2}\right)^{-\frac{\nu+1}{2}}$$



- ν sets the tails. $\nu \to \infty$ gives Gaussian. $\nu = 1$ gives Cauchy.
- s sets the width. $\mathrm{Var}(w) = s^2\nu/(\nu-2)$, finite only for $\nu > 2$.
- It is a Gaussian with a random variance (Lange et al., 1989).

Student-t noise: results

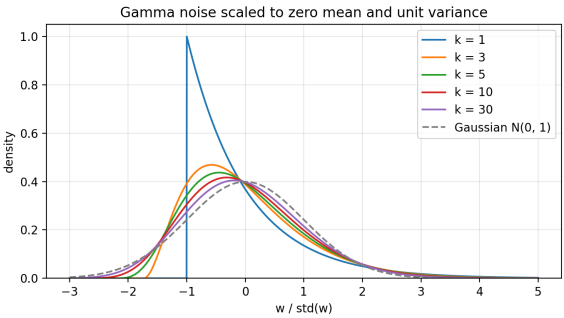


$$\mathrm{CRLB} = \frac{\nu+3}{\nu+1} \cdot \frac{s^2}{N}$$

$$\frac{\mathrm{Var}(\bar{x})}{\mathrm{CRLB}} = \frac{\nu(\nu+1)}{(\nu-2)(\nu+3)}$$

- Heavy tails make the mean much worse (leftwards in the plot).
- The median stays between 1.2 and 1.5.
- They are equal at $\nu \approx 4.68$.
- The t-MLE is a weighted mean with weight $(\nu+1)/(\nu+r^2)$. It reaches the CRLB.

Skewed noise: the gamma family



Noise: $w = Y - k\theta$,
$Y \sim \text{Gamma}(k, \theta)$.

- Zero mean, skewness $2/\sqrt{k}$.
- $k = 1$: exponential. Large k: almost Gaussian.

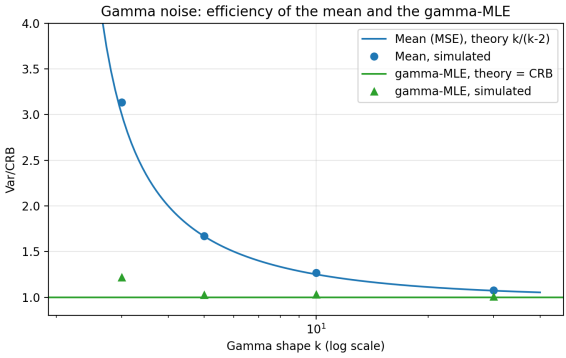
CRLB (valid for $k > 2$):

$$\text{CRLB}_\gamma = \frac{(k-2)\,\theta^2}{N}$$

Mean (MSE):

$$\frac{\text{Var}(\bar{x})}{\text{CRLB}_\gamma} = \frac{k}{k-2}$$

Gamma noise: simulation results



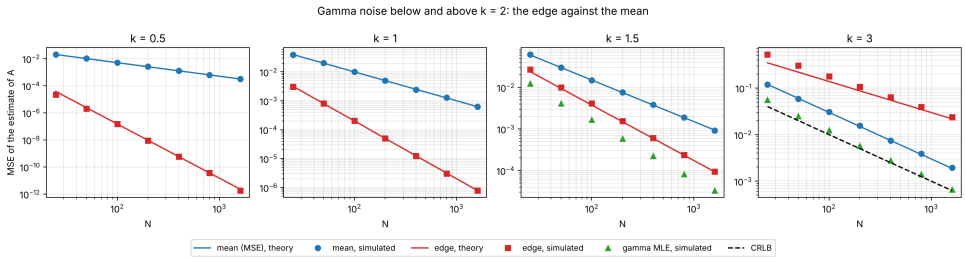
$A = 1$, $\theta = 1$, $N = 100$, 5000 trials.

- **Mean (MSE):** unbiased, but wastes data. $3\times$ the CRLB at $k = 3$.
- **Gamma MLE:** approaches the CRLB slowly (1.18 at $N = 100$, 1.04 at $N = 1000$).
- **Median (MAE):** lands at about $A - \theta/3 \approx 0.67$ (0.69 at $k = 1$), not $A = 1$, for every k. More data does not fix it.

Below $k = 2$: no CRLB, but the edge is informative

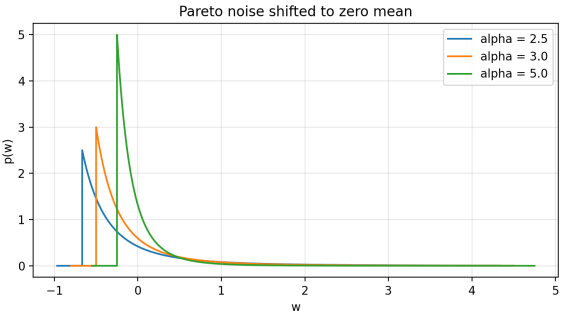
The data have a hard left edge at $A - k\theta$. **Edge estimator:** $\hat{A}_{\text{edge}} = \min_n x[n] + k\theta$ (this *is* the gamma MLE for $k \leq 1$).

$$\text{MSE}(\hat{A}_{\text{edge}}) \approx \theta^2 \, \Gamma\left(1 + \tfrac{2}{k}\right) \Gamma(k+1)^{2/k} \, N^{-2/k} \qquad \text{vs} \qquad \text{MSE}(\bar{x}) = \frac{k\theta^2}{N}$$



- $k < 2$: the edge wins, at rate $N^{-2/k}$. At $N = 100$, $k = 1$: **50**$\times$ better than the mean.
- $k > 2$: the mean wins. $k = 2$ is exactly where the Fisher information becomes infinite.

When the bound breaks: Pareto noise



$P \sim \text{Pareto}(\alpha)$: $\quad p(p) = \dfrac{\alpha}{p^{\alpha+1}}, \quad p \geq 1.$

Noise: $w = P - \dfrac{\alpha}{\alpha - 1}$ (zero mean).

- Heavy power-law tail: $E[P^r]$ finite only for $r < \alpha$.
- **New feature:** the density does not start at zero. It **jumps** from 0 to α at the left edge.
- The edge, $A - \frac{1}{\alpha-1}$, moves with A.
- Used for heavy-tailed radar sea clutter.

Why the CRLB does not apply

Score of one sample, with $u =$ the value of P:

$$s(x;A) = \frac{\partial}{\partial A} \ln p(x;A) = \frac{\alpha+1}{u}, \qquad E[s] = (\alpha+1)\, E\left[\frac{1}{P}\right] = \alpha \;\neq\; 0.$$

- The CRLB needs $E[s] = 0$. It fails here.
- Reason: the support moves with A. Differentiating $\int p(x;A)\,dx = 1$ gives an extra term equal to the **density at the edge**. It is zero only if the density is zero there. Here it is α.
- Computing the formula anyway gives a **naive CRLB**:

$$\text{naive CRLB} = \frac{\alpha+2}{N\,\alpha(\alpha+1)^2} \qquad \text{(falls like } 1/N\text{)}$$

- We will see an estimator beat it by $100\times$, so it is not a valid bound.

Three estimators, and a bound that still holds

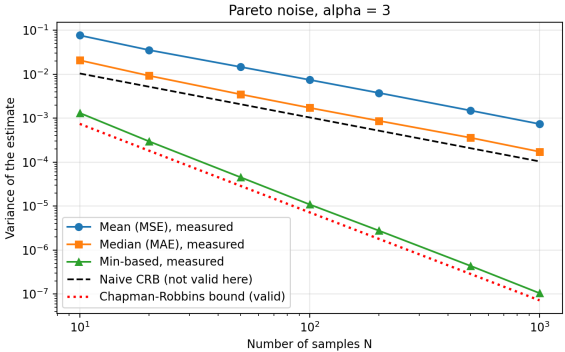
Estimator	Variance	Rate
Mean (MSE)	$\dfrac{\alpha}{(\alpha-1)^2(\alpha-2)\,N} \quad (\alpha > 2)$	$1/N$
Median (MAE)	$\approx \dfrac{2^{2/\alpha}}{N\alpha^2}$, biased: goes to $A + 2^{1/\alpha} - \frac{\alpha}{\alpha-1}$	$1/N$
Min-based	$\dfrac{N\alpha}{(N\alpha-1)^2(N\alpha-2)} \approx \dfrac{1}{(N\alpha)^2}$	$\mathbf{1/N^2}$

- The minimum of N Pareto(α) samples is Pareto($N\alpha$), so we can remove its bias exactly: $\hat{A}_{\min} = \min_n x[n] + \frac{\alpha}{\alpha-1} - \frac{N\alpha}{N\alpha-1}$.
- **Chapman–Robbins bound** (no regularity needed): compare A with a nearby $A + h$,

$$\mathrm{Var}(\hat{A}) \geq \sup_{h>0} \frac{h^2}{E_1(h)^N - 1}.$$

If the data cannot tell A from $A + h$, no estimator can be precise. As $h \to 0$ it becomes the CRLB when that is valid.

Pareto noise: simulation results

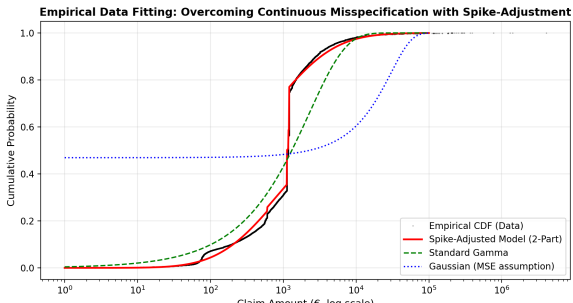


$A = 1$, $\alpha = 3$, 5000 trials.

- All measured variances match the formulas.
- At $N = 100$: min-based 1.16×10^{-5} vs naive CRLB 1.04×10^{-3}: **almost** $100\times$ **below** the "bound".
- Slopes: mean, median, naive CRLB -1; min-based -2.04.
- Chapman–Robbins: slope -2.01, just below the min-based estimator. **Valid and nearly tight.**

Takeaway: a hard edge is very informative. MSE and MAE ignore it, so both are far from the best estimator.

Real data: French motor insurance claims

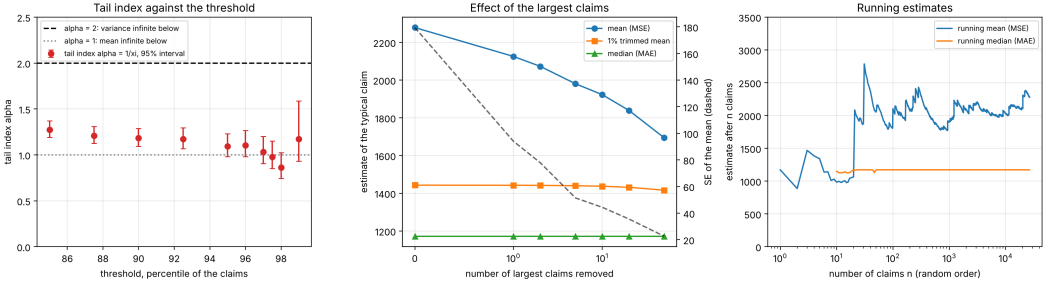


26,639 claim amounts (`freMTPL2sev`). Noise unknown: we fit it.

- Mean 2279, median 1172.
- 42% of claims sit on 5 fixed amounts. A two-part model (spikes + lognormal) fits them.
- Best gamma fit: $k = 0.69 < 1$. No CRLB exists.
- MSE vs MAE is a choice of *what* to estimate, not of efficiency.

Real data: how precise is the mean?

SE of the mean: gamma model 16.8 vs sample MCRB 179.5 $\approx$ bootstrap 177.0 $\Rightarrow$ the wrong model is 10.7$\times$ **overconfident**.



- Tail index α between 0.86 and 1.28 for every threshold: $\alpha < 2$, so the variance is **infinite**.
- Remove 1 claim of 26,639: the mean moves by 153 and its SE halves. The median never moves.
- **Takeaway:** on these data MAE gives a stable answer. MSE does not, and no error bar on it can be trusted.

Neural networks: the loss is still a noise model

From one number A to a whole function:

$$y = f^*(x) + w, \qquad f^*(x) = \sin(2\pi x) + 0.5x, \qquad x \sim \mathcal{U}[0, 1].$$

- Network f_θ: MLP, 3 hidden layers of width 64, GELU, AdamW, 400 epochs. $N = 250$, 25 independent trials.
- Losses compared: MSE (Gaussian), MAE (Laplace), Huber, the matched likelihood, and **RCOS**, which picks the loss from the data.
- Noise: Gaussian ($\sigma = 0.3$), centered gamma ($k = 3$, skewed), Student-t ($\nu = 3$, heavy tails).

Error measure over the 25 trained networks, with $\bar{f}$ their average:

$$\text{Total risk} = \underbrace{\tfrac{1}{K}\textstyle\sum_k \left(\bar{f}(x_k) - f^*(x_k)\right)^2}_{\text{Bias}^2} + \underbrace{\tfrac{1}{K}\textstyle\sum_k \tfrac{1}{M}\textstyle\sum_m \left(f^{(m)}(x_k) - \bar{f}(x_k)\right)^2}_{\text{Variance}}$$

The same split as in the DC case: a wrong target shows up as bias, a wasteful loss as variance.

RCOS: choosing the loss from the residuals

1. Split the training data: 75% to fit, 25% to calibrate. The test set is never touched.
2. Train a pilot network with Huber loss, and compute residuals r_i on the calibration part.
3. Measure skewness $\hat{\gamma}_1$, excess kurtosis $\hat{\kappa}$ and the tail ratio $\hat{\sigma}/\hat{\sigma}_{\mathrm{MAD}}$.
4. Choose the loss:
   - $|\hat{\gamma}_1| \geq 0.45$ (skewed) $\Rightarrow$ gamma likelihood, with $\hat{k} = 4/\hat{\gamma}_1^2$
   - $\hat{\kappa} \geq 0.8$ or tail ratio ≥ 1.22 (heavy tails) $\Rightarrow$ Student-t likelihood, $\hat{\nu} = \max(2.1,\, 4 + 6/\hat{\kappa})$
   - otherwise $\Rightarrow$ MSE
5. Retrain on all the training data with the chosen loss.

The rules use the moment formulas of the earlier sections: gamma skewness $2/\sqrt{k}$, Student-t excess kurtosis $6/(\nu - 4)$.

Neural networks: simulation results

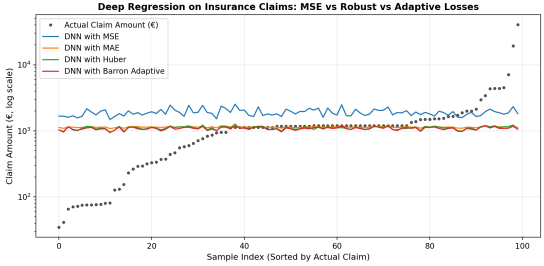
Total risk (Bias^2)	Gaussian	Gamma ($k = 3$)	Student-t ($\nu = 3$)
MSE	0.068 (0.030)	0.072 (0.033)	0.079 (0.040)
MAE	0.098 (0.056)	0.096 (**0.052**)	0.089 (0.047)
Huber	0.028 (0.009)	0.047 (0.017)	0.029 (0.007)
RCOS	0.083 (0.039)	0.082 (0.043)	**0.066** (0.022)

$N = 250$, 25 trials.

- **Gaussian:** RCOS chose MSE in 25 of 25 trials, so it does not raise false alarms.
- **Student-t:** RCOS has 16% less risk than MSE (0.066 vs 0.079).
- **Gamma:** MAE has the largest bias (0.052 vs 0.033 for MSE). It fits the conditional median, shifted from the mean as in the DC case.
- Huber, the pilot loss, has the lowest risk in all three regimes.

Neural networks on the insurance claims

Predict each claim from the policy features (driver age, vehicle, area, . . .). Test set: 20% of the policies.



Loss	Typical error	Typical prediction
MSE	876	1869
MAE	273	1143
Huber	269	1115
Barron	268	1094

(median absolute error and median prediction, euros)

- MSE predicts a typical claim of 1869, pulled up by the largest claims, with about 3× the typical error.
- The robust losses stay near the 1172 median.
- Same lesson as the DC case: MSE estimates the mean, which the tail dominates.

Thank you. Questions?